

Project Report

Title: Multiple Sequence Alignment (MSA) and Comparative Analysis
Bioinformatics (2025)

1. Introduction

Multiple Sequence Alignment (MSA) is a foundational technique in bioinformatics. It allows us to:

- Identify conserved regions in protein or DNA sequences
- Study evolutionary relationships across species
- Analyze protein function and structure

In this project, a custom MSA method was developed to align Hemoglobin subunit beta (HBB) sequences from multiple vertebrate species. The method's performance was evaluated against Clustal Omega, a widely used MSA tool.

2. Dataset Description

2.1 Training Dataset

The training dataset (**proteins_train.fasta**) includes HBB sequences from:

- *Homo sapiens* (Human)
- *Mus musculus* (Mouse)
- *Rattus norvegicus* (Rat)
- *Canis lupus familiaris* (Dog)
- *Bos taurus* (Cow)
- *Gallus gallus* (Chicken)
- *Xenopus laevis* (African clawed frog)
- *Danio rerio* (Zebrafish)

These sequences were used to optimize parameters like gap penalties and to construct guide trees for progressive alignment.

2.2 Test Dataset

The test dataset (**proteins_test.fasta**) contains previously unseen sequences:

- *Bos taurus* (Cow)
- *Sus scrofa* (Pig)
- *Monodelphis domestica* (Opossum)
- *Oryzias latipes* (Medaka fish)

This set was used exclusively to evaluate alignment accuracy and compare with Clustal Omega.

3. Methodology

3.1 Algorithm Workflow

1. **Preprocessing:**
 - Load sequences from FASTA files using Biopython.
 - Validate lengths and convert sequences to uppercase.
2. **Pairwise Alignment:**
 - Perform global alignment with Needleman-Wunsch algorithm.
 - Use **BLOSUM62** substitution matrix and affine gap penalties.
3. **Distance Matrix:**
 - Compute pairwise distances as $1 - \text{Sequence Identity}$.
 - Build a symmetric distance matrix.
4. **Guide Tree Construction:**
 - Apply UPGMA clustering to generate a guide tree for progressive alignment.
5. **Progressive MSA:**
 - Align sequences according to the guide tree.
 - Two alignment strategies:
 1. Using the first sequence as a reference.
 2. Using a consensus sequence as a reference.
6. **Output:**
 - Save aligned sequences to FASTA files: custom_msa_first.fasta and custom_msa_consensus.fasta.

- Compute alignment statistics: conserved columns, alignment length, gap fraction, and column-wise identity.

3.2 Scoring Scheme and Parameters

Parameter	Value
Substitution matrix	BLOSUM62
Gap opening penalty	6
Gap extension penalty	0.5
Distance metric	1 – Identity

- Gap penalties were optimized using the training dataset to maximize conservation while minimizing gaps.

4. Parameter Optimization Results

- **Best gap opening penalty: 6**
- **Best gap extension penalty: 0.5**

These values gave the highest number of fully conserved columns and lowest gap fraction.

5. Custom MSA Results

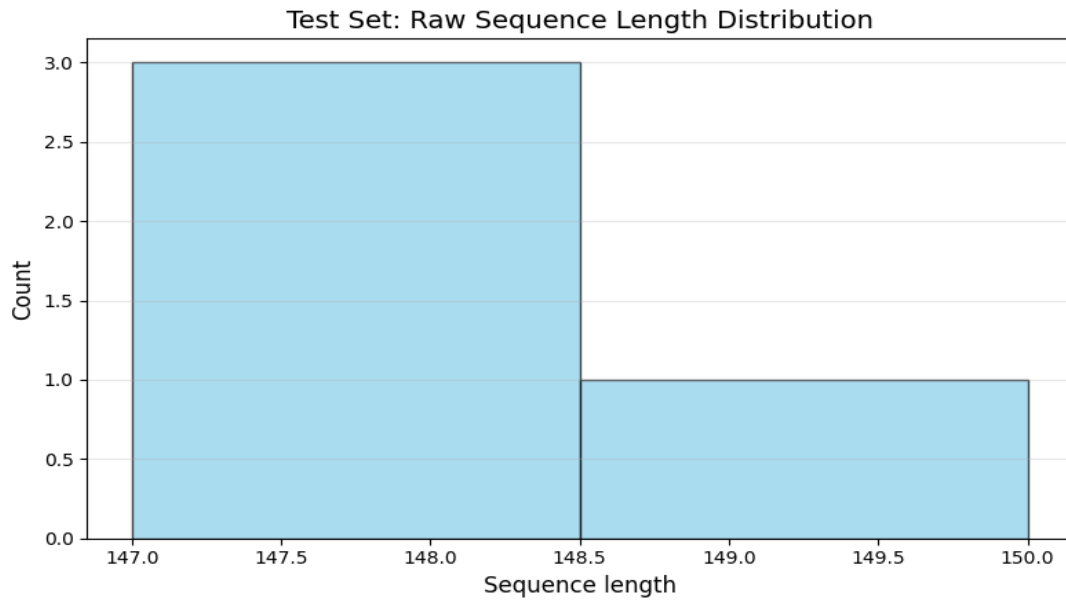
5.1 Alignment Statistics

Metric	First Seq	Consensus
Alignment length	150	150
Fully conserved columns	119	119
Total gap fraction	0.0150	0.0150
Average gaps per sequence	2.25	2.25

- About 79.3% of columns are fully conserved, indicating high similarity across sequences.
- Both strategies (first sequence vs. consensus) produced similar results.

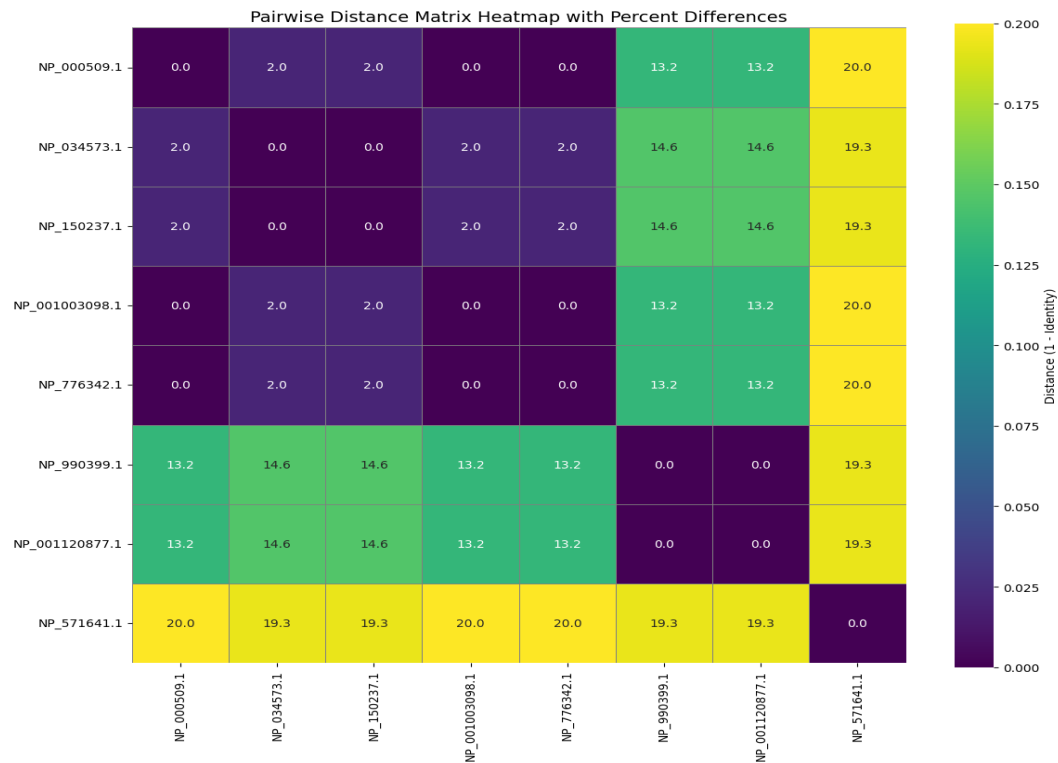
5.2 Visualization and Interpretation

1. Raw Sequence Lengths



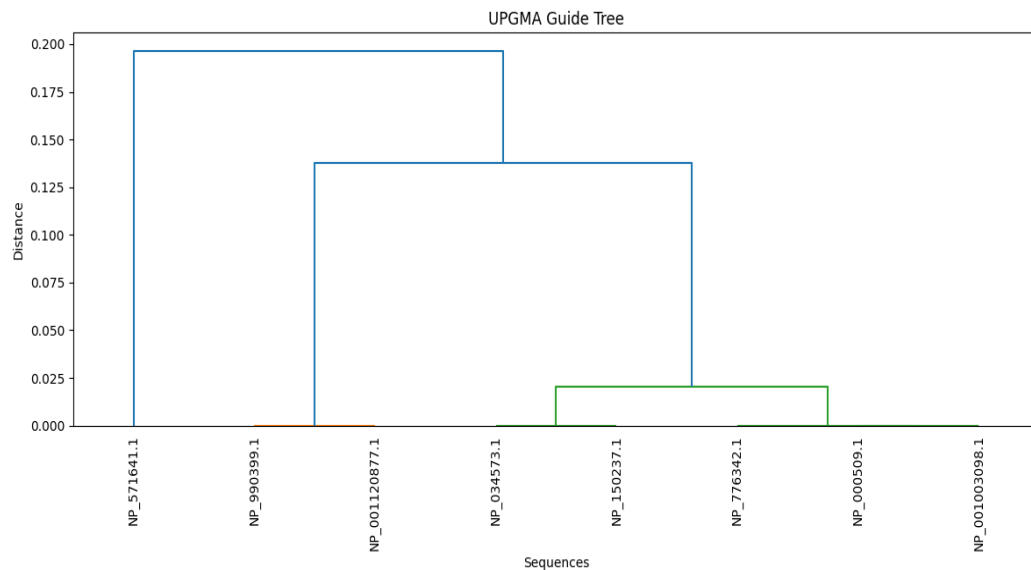
- Test sequences range from 147–150 amino acids (mean = 147.8).
- Minor length differences justify global alignment.

2. Pairwise Distance Matrix with Percent Differences



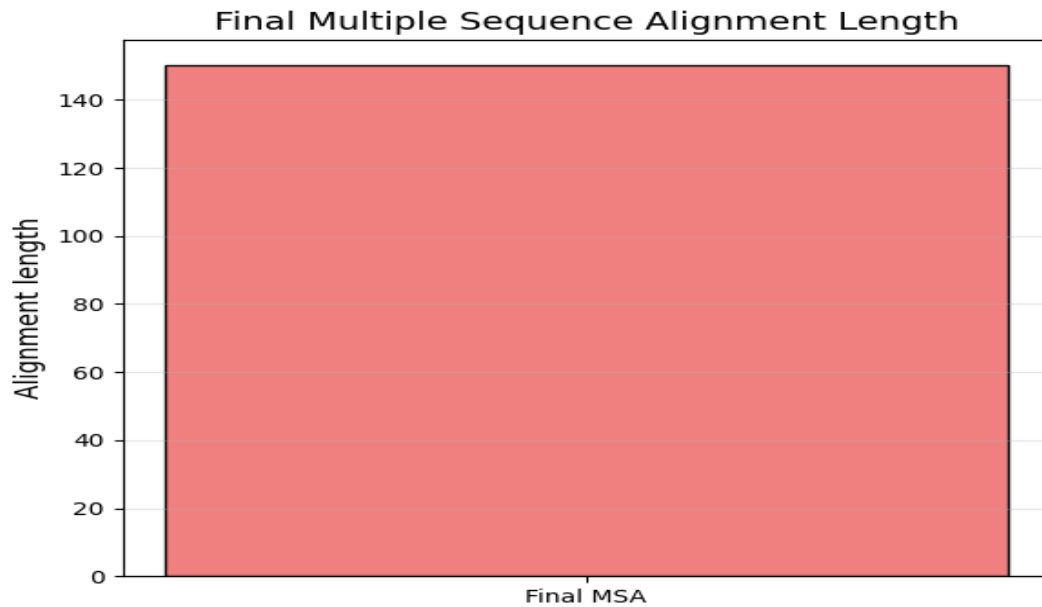
- Displays pairwise percent differences between all sequences.
- Highlights which sequences are more similar or divergent.

3. UPGMA Guide Tree



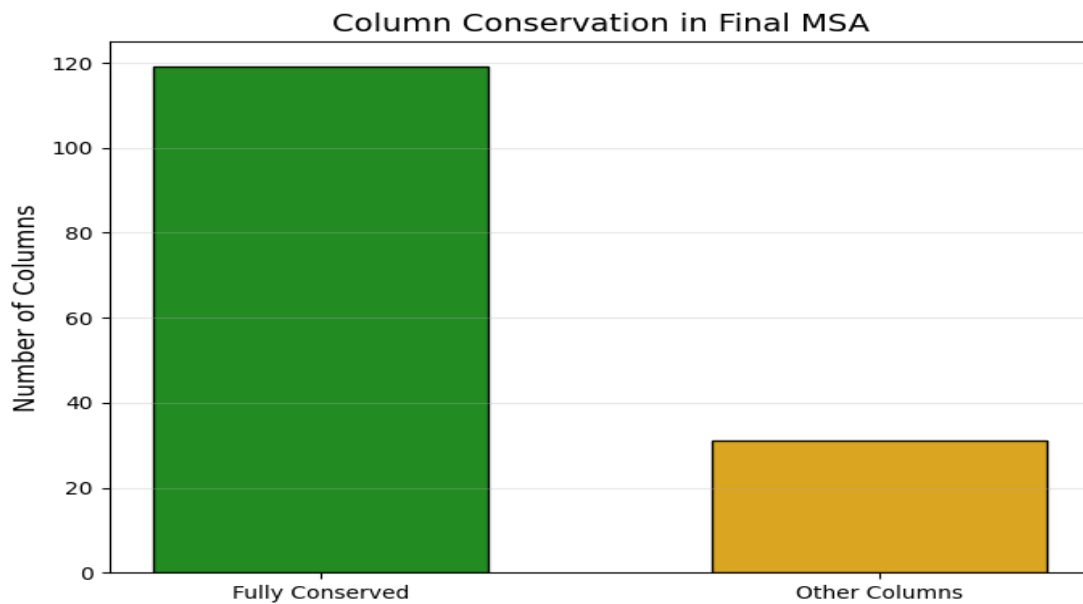
- Generated from the pairwise distance matrix.
- Shows the hierarchical clustering of sequences for progressive alignment.

4. Final Alignment Length



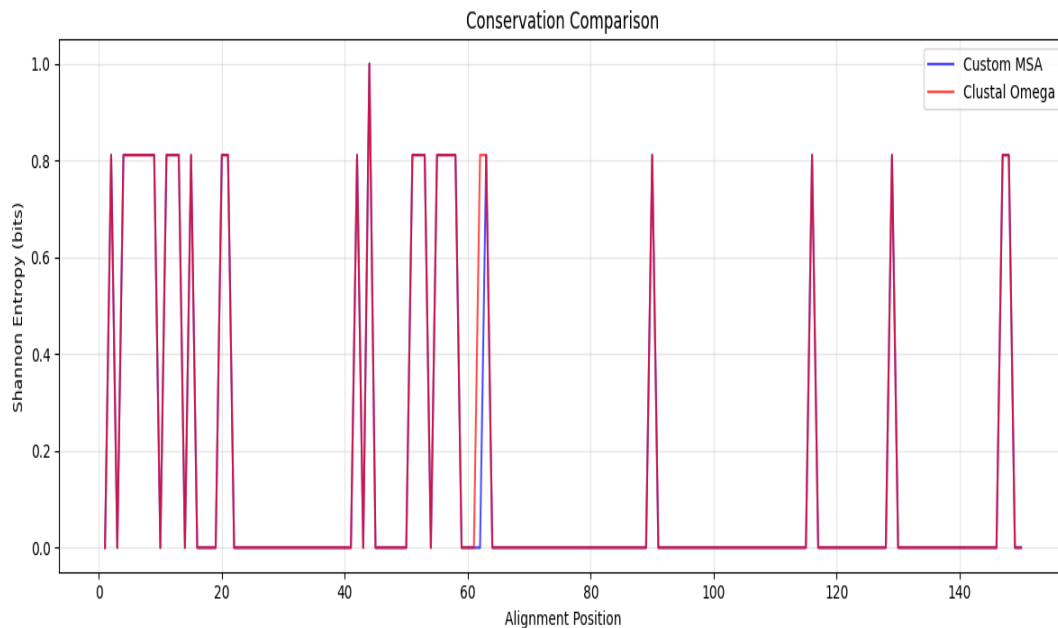
- Alignment length standardized to 150 after introducing gaps.

5. Conserved Columns



- 119 columns fully conserved (~79.3%).
- Suggests strong evolutionary conservation among HBB proteins.

6. Conservation Comparison (Shannon Entropy)



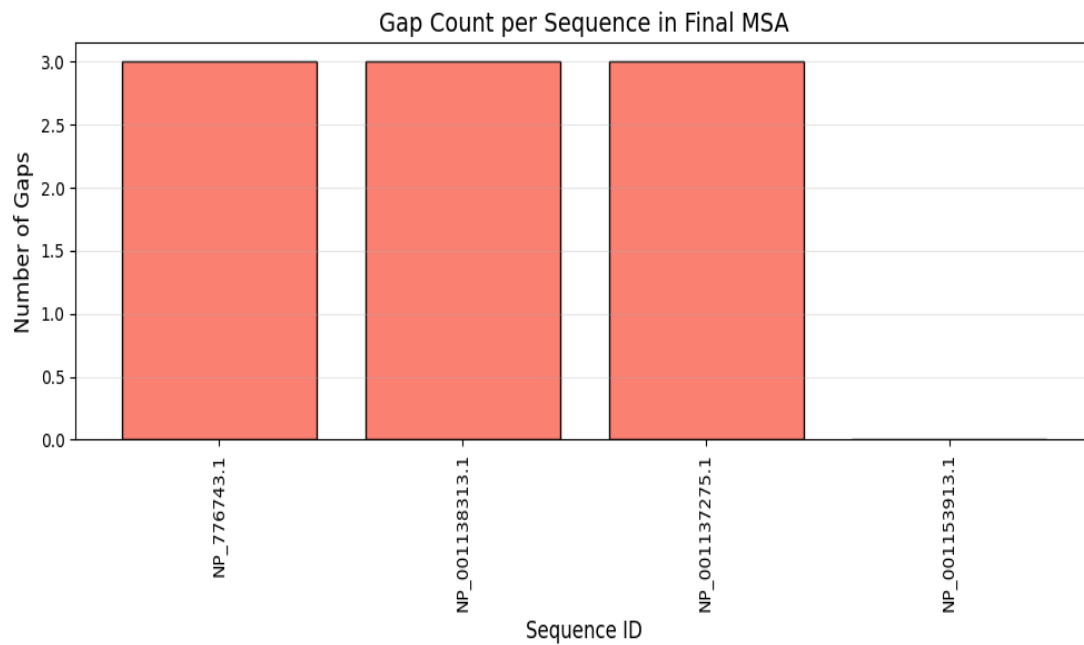
- Compares column-wise conservation (via Shannon entropy) between Custom MSA and Clustal Omega.
- Lower entropy = higher conservation.

7. Column-wise Identity



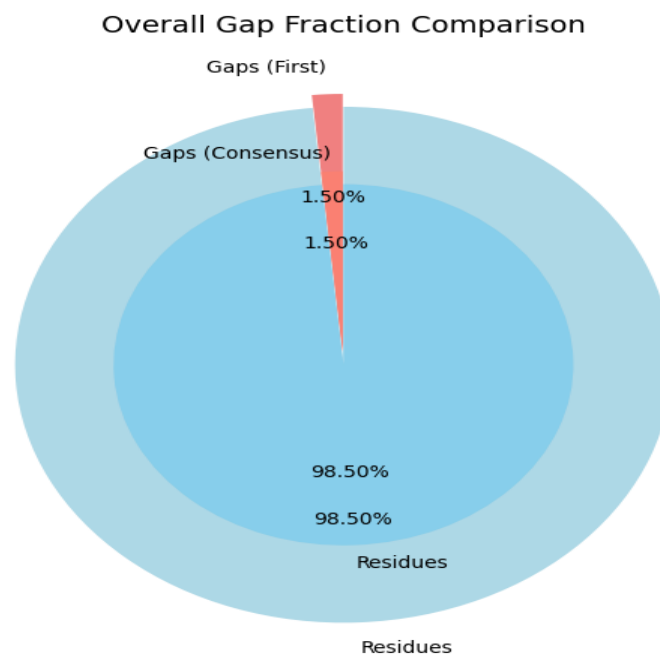
- Measures similarity between Custom MSA and Clustal Omega on a per-column basis.
- Column-wise identity = 98.67%.

8. Gaps per Sequence



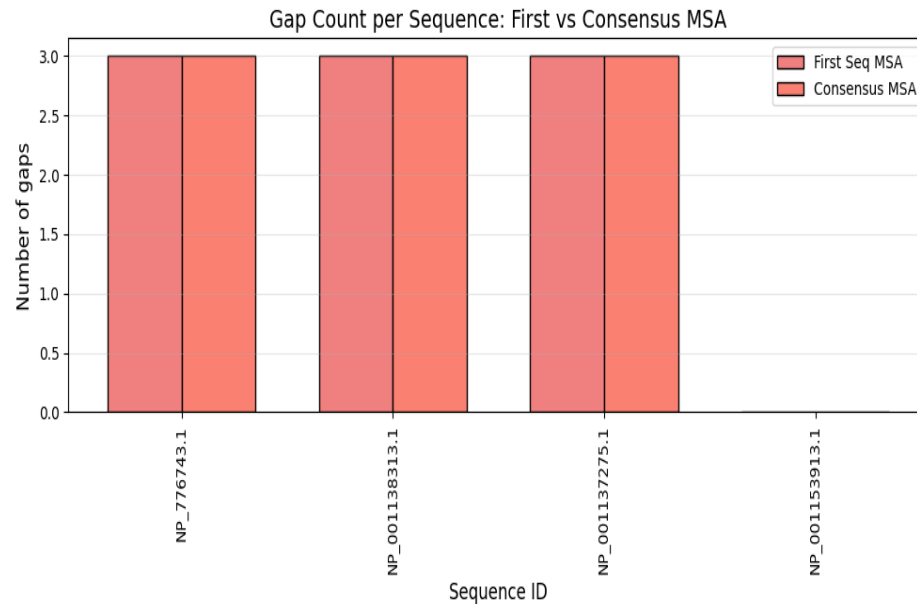
- Average of 2.25 gaps per sequence.
- Gaps are minimal and evenly distributed.

9. Overall Gap Fraction Comparison



- First-sequence and consensus strategies give similar gap fractions (~0.015).

10. Gap per Sequence Comparison



- Confirms low insertion/deletion events across sequences.

6. Comparison with Clustal Omega

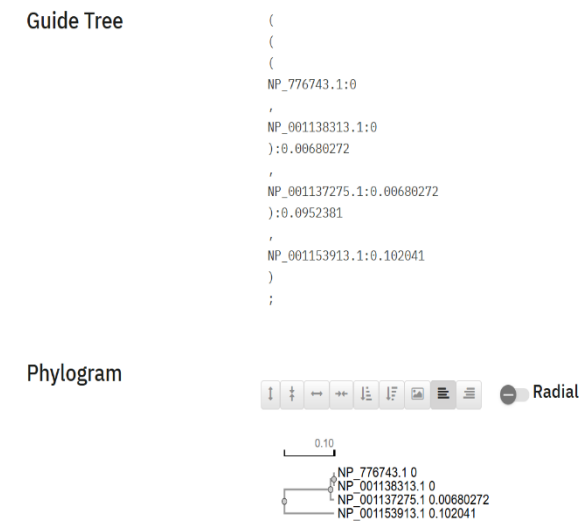
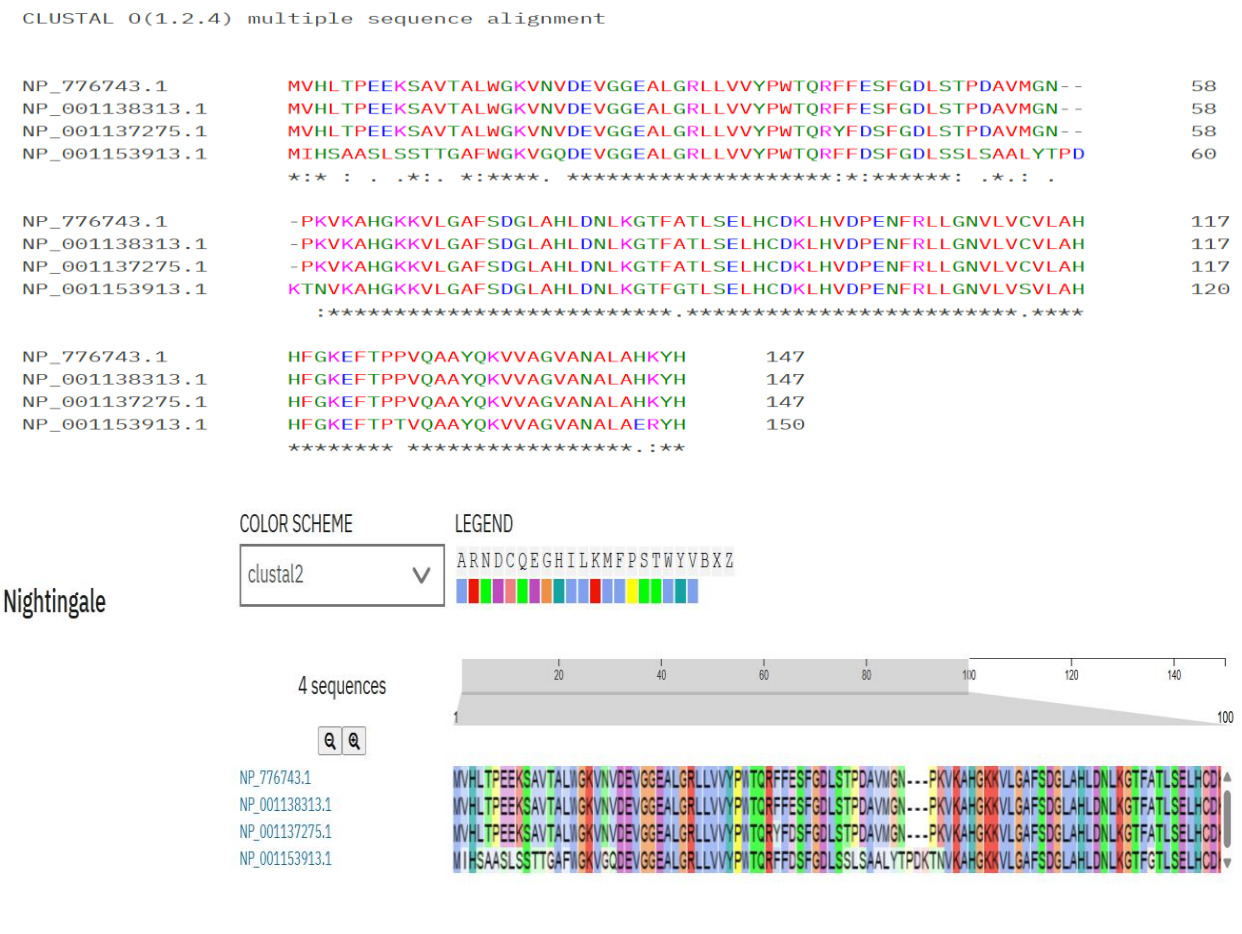
- Clustal Omega alignment performed online on the same test sequences.

METRIC	CUSTOM MSA	CLUSTAL OMEGA
ALIGNMENT LENGTH	150	150
FULLY CONSERVED COLUMNS	119	118
AVG GAPS PER SEQUENCE	2.25	2.25
COLUMN-WISE IDENTITY (%)	98.67	100

Observations:

- Custom MSA and Clustal Omega produce almost identical results.
- Slightly more fully conserved columns in custom MSA (119 vs 118).
- Column-wise identity very high (98.67%), confirming reliable alignment.

Clustal Omega alignment screenshot placeholder:



Phylogenetic Tree

```
(  
(  
NP_776743.1:0.00000,  
NP_001138313.1:0.00000)  
:0.00600,  
NP_001137275.1:0.00600,  
NP_001153913.1:0.18367);
```

Phylogram



```
└──  
└── NP_776743.1 0.00000  
└── NP_001138313.1 0.00000  
└── NP_001137275.1 0.00600  
└── NP_001153913.1 0.18367
```

Selected **0** branches with current label

```
... Clustal Omega alignment done.  
  
CUSTOM MSA STATISTICS  
Alignment length: 150  
Fully conserved columns: 119 (79.33%)  
Average gaps per sequence: 2.25  
  
COMPARISON TABLE  

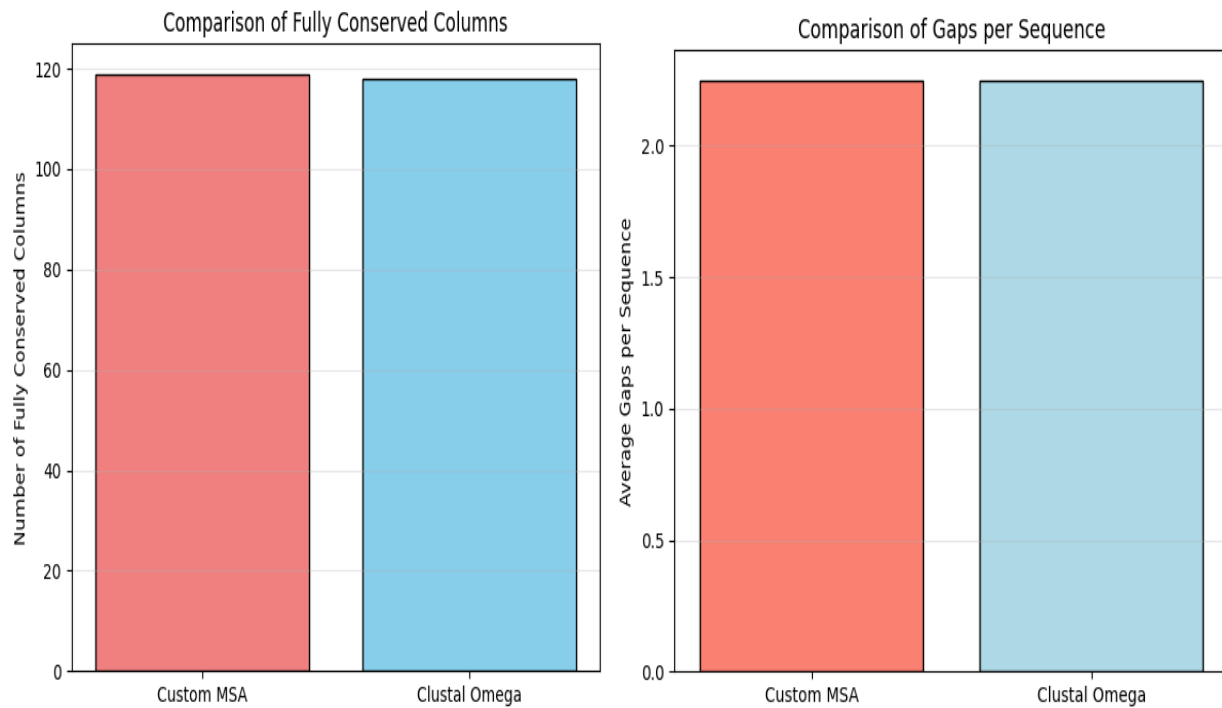

|                         | Metric Custom MSA | Clustal Omega |
|-------------------------|-------------------|---------------|
| Alignment Length        | 150               | 150           |
| Fully Conserved Columns | 119               | 118           |
| Avg Gaps per Sequence   | 2.25              | 2.25          |

  
Column-wise identity: 98.67%
```

7. Conclusion

- Custom progressive MSA method accurately aligns HBB sequences.
- Parameter optimization improves conservation without introducing excessive gaps.
- Comparison with Clustal Omega shows:
 - Similar alignment length and gap placement
 - High percentage of conserved columns
 - Robust overall column-wise identity
- Consensus-based reference slightly enhances conservation.

Final Statistics Summary:



Metric	Value
Final alignment length	150
Fully conserved columns	119 (79.33%)
Average gaps per sequence	2.25
Overall gap fraction	0.0150
Column-wise identity vs Clustal Ω	98.67%